

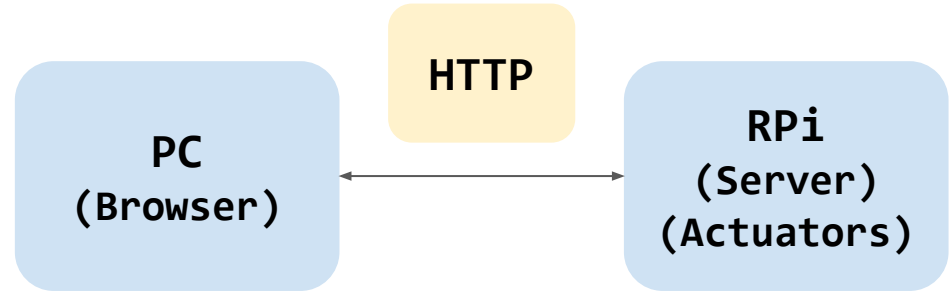
# Hand-Controlled Catapult Car

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# **Project Preview**

# Project title

- A Mobile Catapult Controlled by **Real-Time Hand Gestures**.
- the browser handles image capture and gesture recognition
- the Raspberry Pi handles hardware control



# Motivation

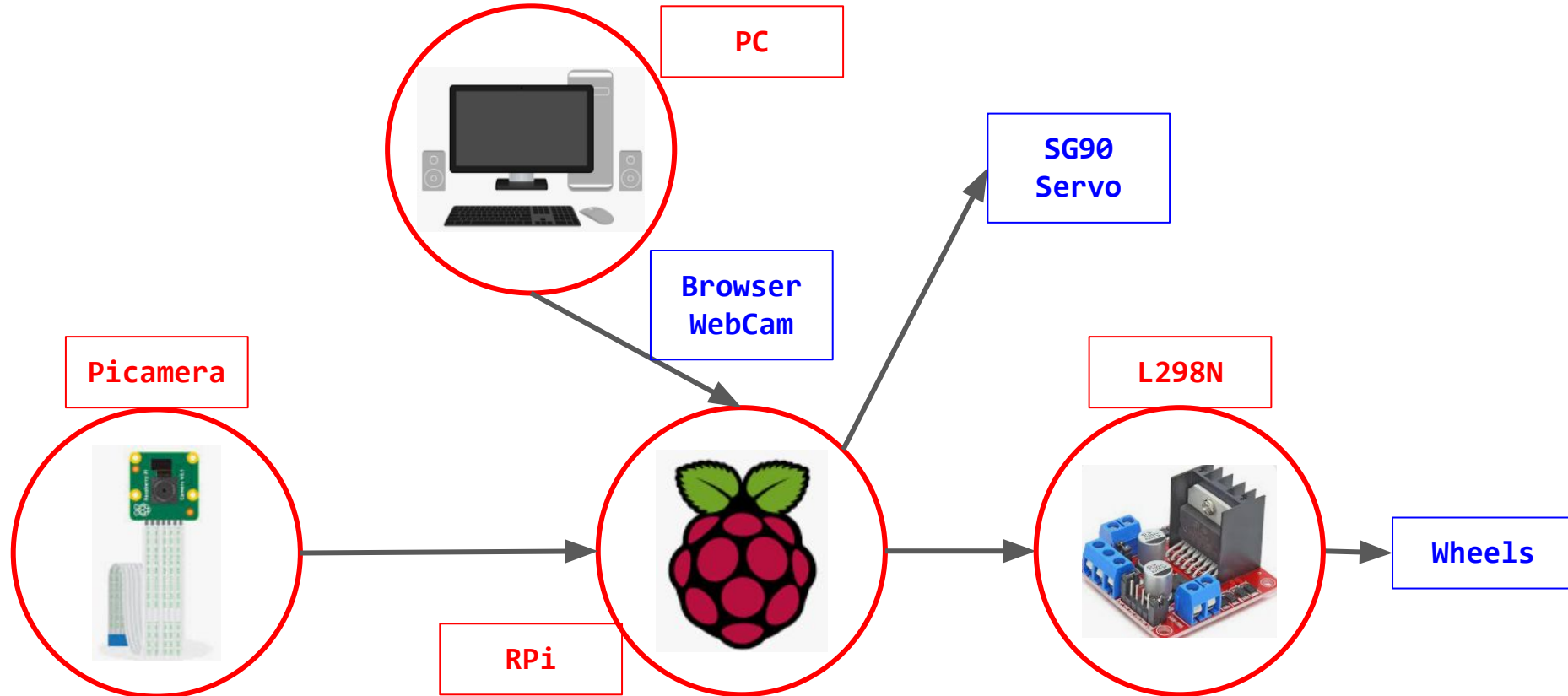
Motivated by the in-class assignment from the course:

## Intro to Multimedia

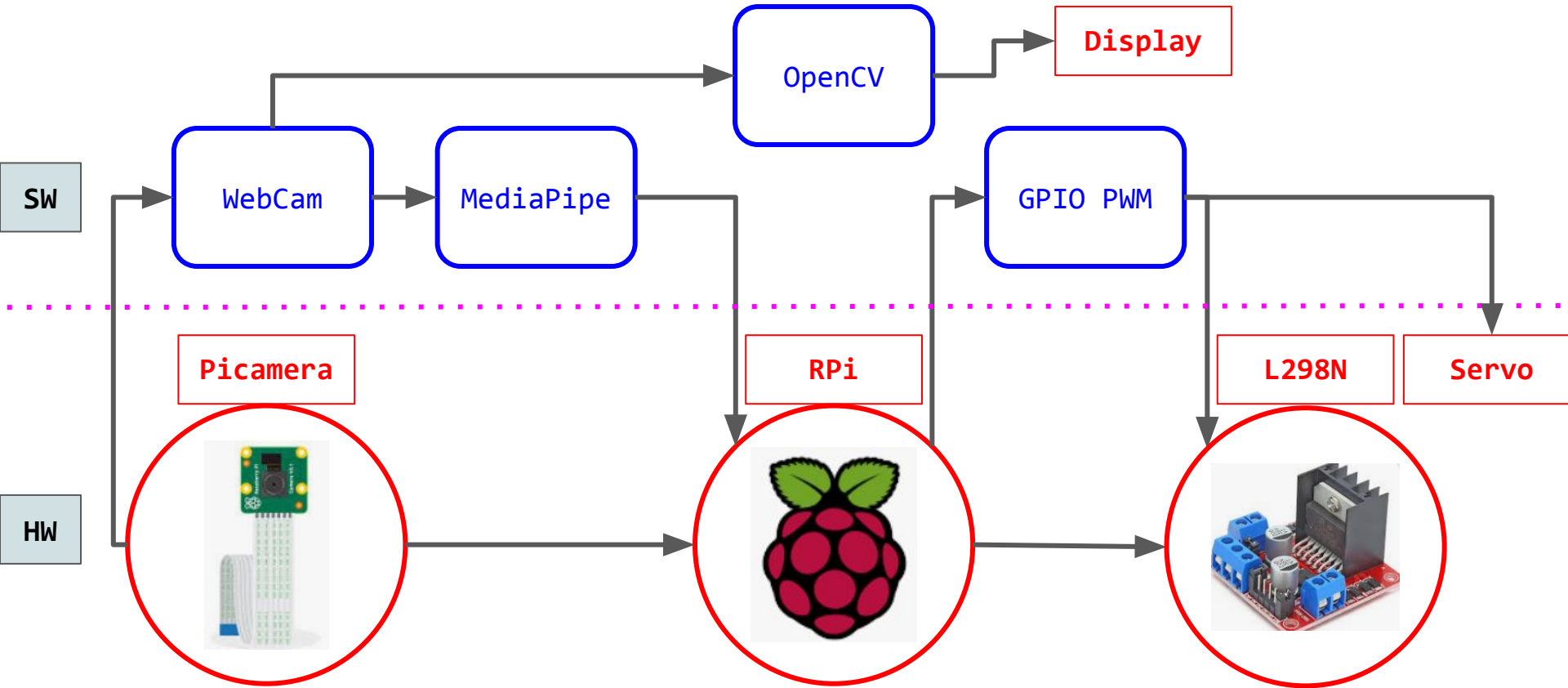
- Integrate the knowledge from two classes.
- Demonstrates a complete input-output chain using Raspberry Pi.



# Architecture

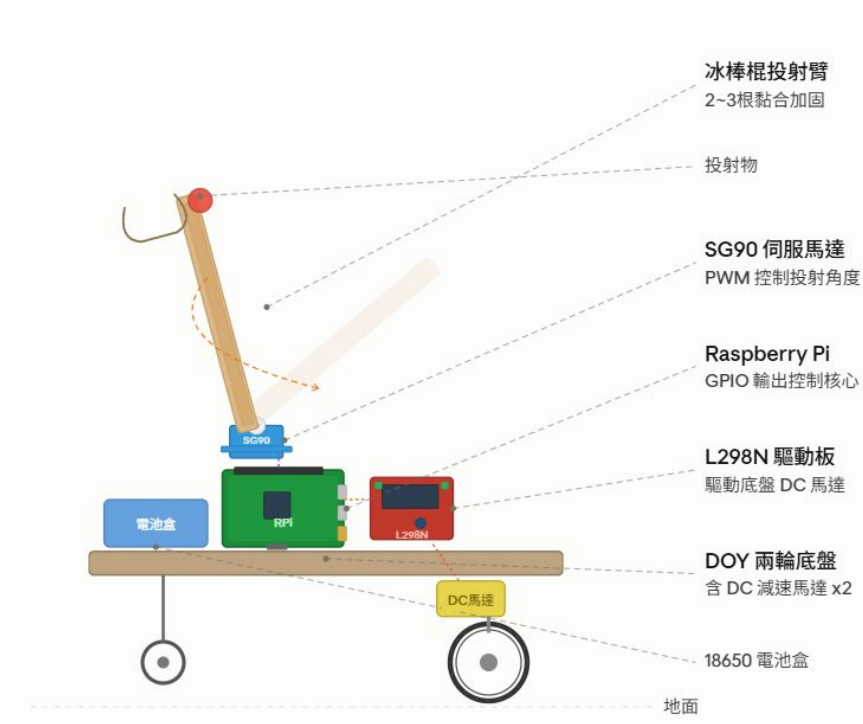


# SW & HW



# Components

1. L298N電機驅動板模組 \$40
2. DIY兩輪智能小車底盤 \$125
3. SG90 Servo\$32
4. 冰棒棍\$0
5. 電池\*4\$0
6. 5V3A行動電源 \$0



# Control

## Hand-Control Mode

- Fire: Controlled by gesture
- Direction: Controlled by wheel
- Speed: Controlled by lever
- Differential steering



Fire



steer



Speed



# **The Project**

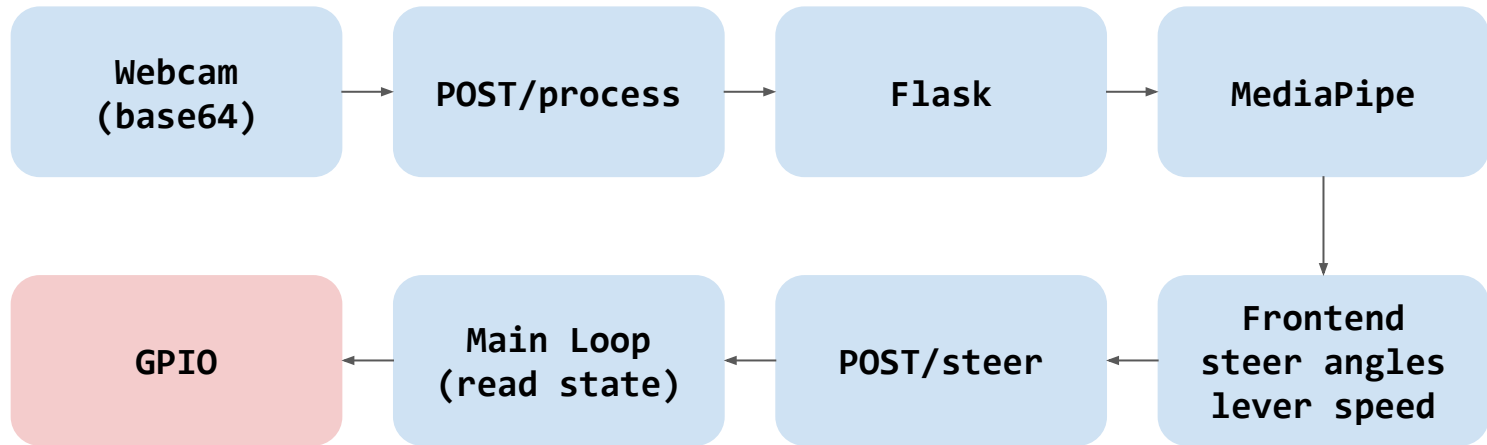
# Program Structure

1. Hand Control
2. SG90 (Fire)
3. Wheels
4. Templates/Web Server

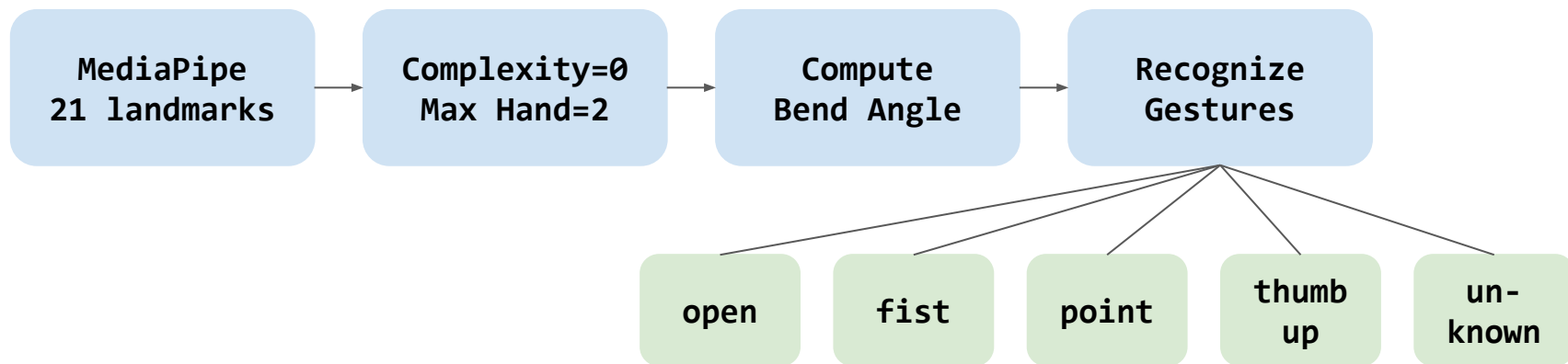
Main

```
— hand_ctrl.py
— main.py
— sg90.py
— templates
  |— index.html
  |— index.html.old
— web_server.py
— web_server.py.old
— wheels.py
```

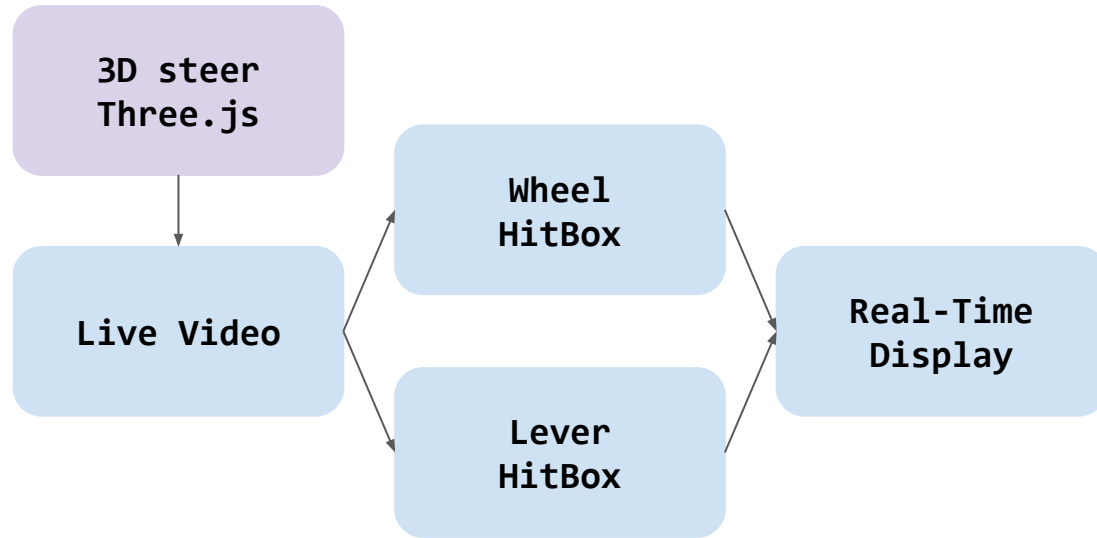
# System Data Flow



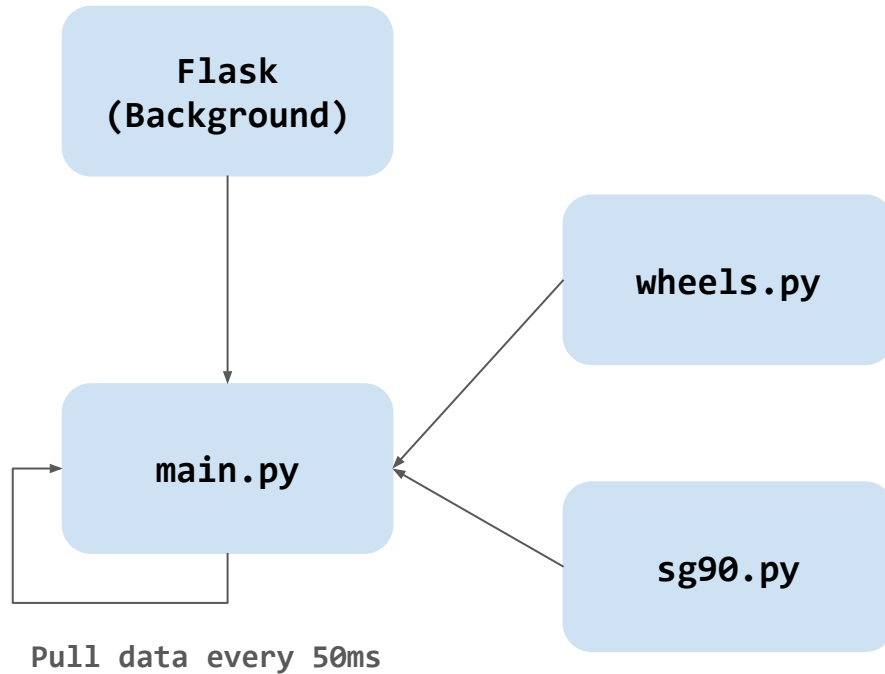
# Hand Recognition



# Control Panel



# Main Loop



# Tasks (With Libs) Summary

| Tasks               | Libs            |
|---------------------|-----------------|
| Gesture Recognition | MediaPipe       |
| Image processing    | OpenCV          |
| Web server          | Flask           |
| 3D steering wheel   | Three.js        |
| Car-view streaming  | Picamera2 MJPEG |
| GPIO / PWM          | GPIO            |
| computation         | NumPy           |

# SW Demo

← → ↻ 172.20.10.4:5000

## Hand Gesture Controller



The interface displays a user wearing headphones and glasses, with a steering wheel overlay on their left hand. The right hand is raised, showing finger positions. A car camera feed is visible on the right.

**Gesture**

Open

**Steering**

-58°  
Steering

**Speed**

0%  
Lever idle

**Hand 1**

| Finger | Color  | Angle |
|--------|--------|-------|
| Thumb  | Red    | 167°  |
| Index  | Orange | 177°  |
| Middle | Yellow | 180°  |
| Ring   | Green  | 176°  |
| Pinky  | Blue   | 175°  |

**Hand 2**

| Finger | Color     | Angle |
|--------|-----------|-------|
| Thumb  | Dark Blue | -     |
| Index  | Dark Blue | -     |
| Middle | Dark Blue | -     |
| Ring   | Dark Blue | -     |
| Pinky  | Dark Blue | -     |

**Status**

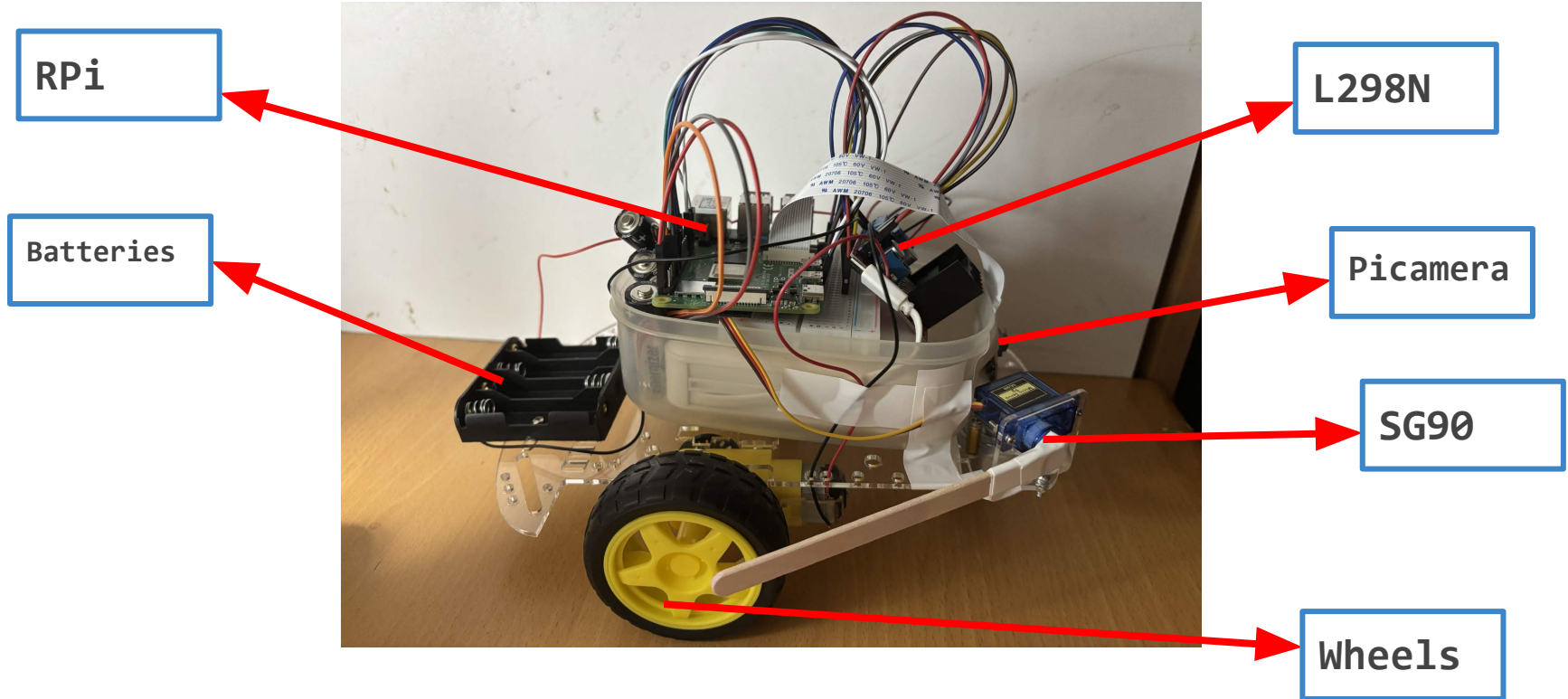
Detecting

Stop

FPS: 5



# HW Demo





**The End**